

Finetuning Large Language Models for Materials Chemistry

Cyprien Bone and Keith T. Butler
Department of Chemistry, University College London

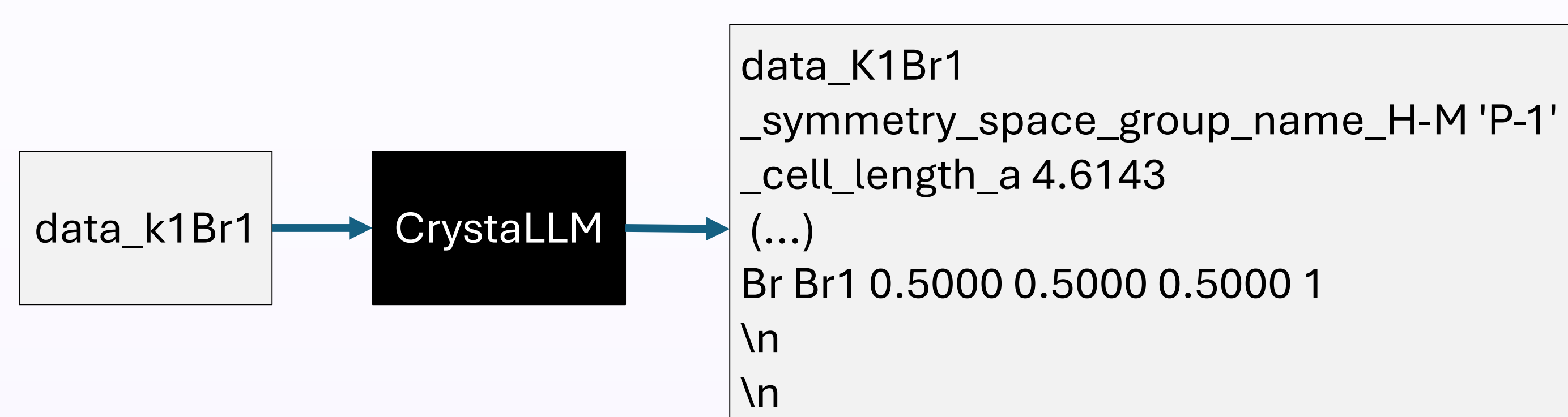


BACKGROUND

In materials chemistry, researchers often possess small to medium specialised datasets, insufficient to train deep learning models from scratch.

Can we use a **Large Language Model's** ability to transfer information learned from large amounts of data for new applications on smaller sets of data?

CrystaLLM¹ is an auto-regressive model based on the GPT-2 architecture (203 million parameters) specifically trained for crystal structure generation.

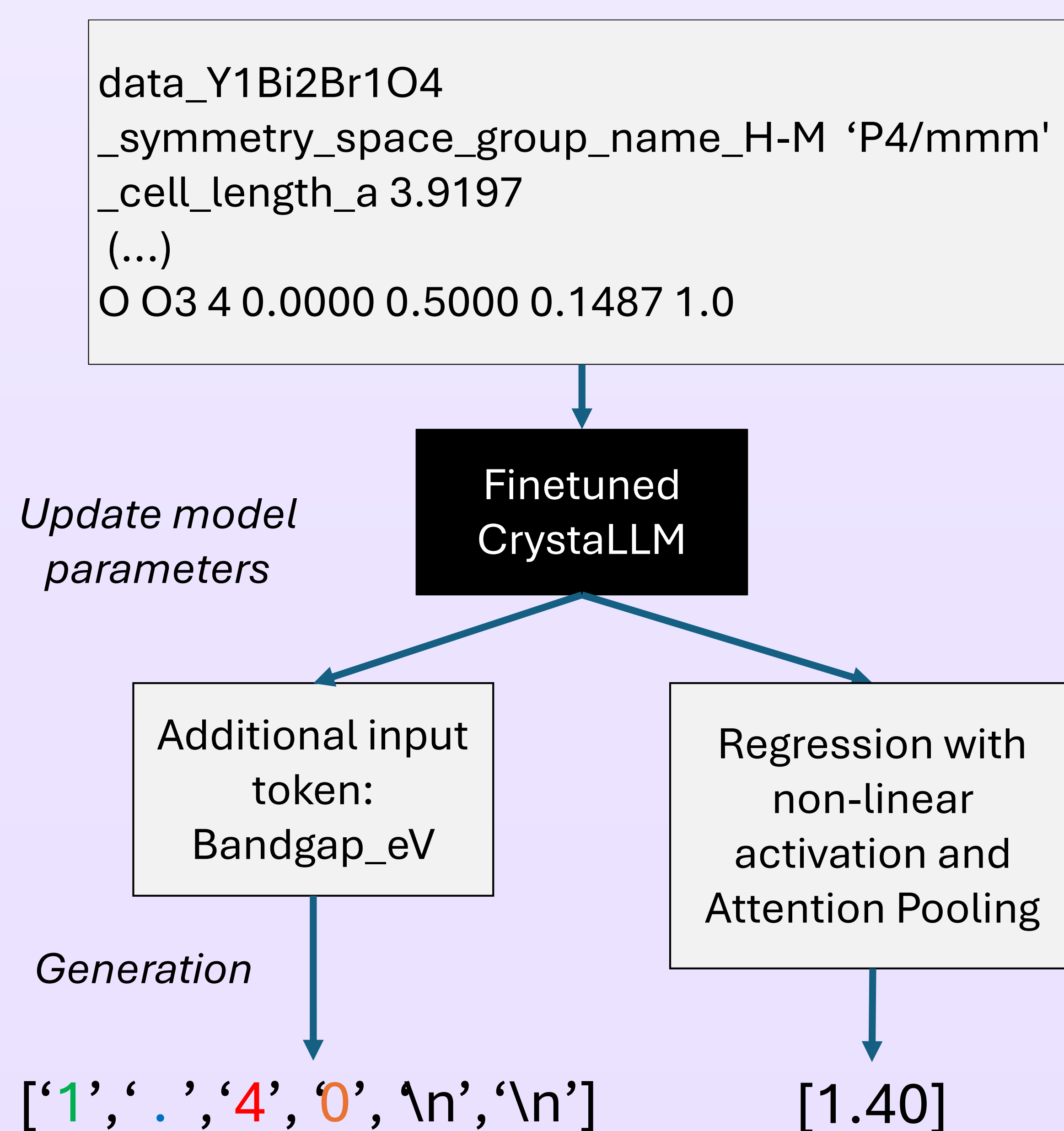


Scheme 1. Generation of a Crystallographic Information File (CIF) using the pre-trained model

OBJECTIVES & METHODS

Explore the importance of target representation to predict material bandgap from ~50K **material project**² crystals:

1. Extend models token vocabulary to allow for bandgap predictions using next token generation
2. Adapt the final layer in the transformer to predict a numerical value using Attention Pooling and sequential linear layers with ReLU activation
3. Evaluate parameter update methods with the goal of reducing computational cost



Scheme 2. Model finetuning workflow

Total time and energy spent developing and training the models:⁶
284.27 and 106.91 kWh (~51 washing cycles)

RESULTS

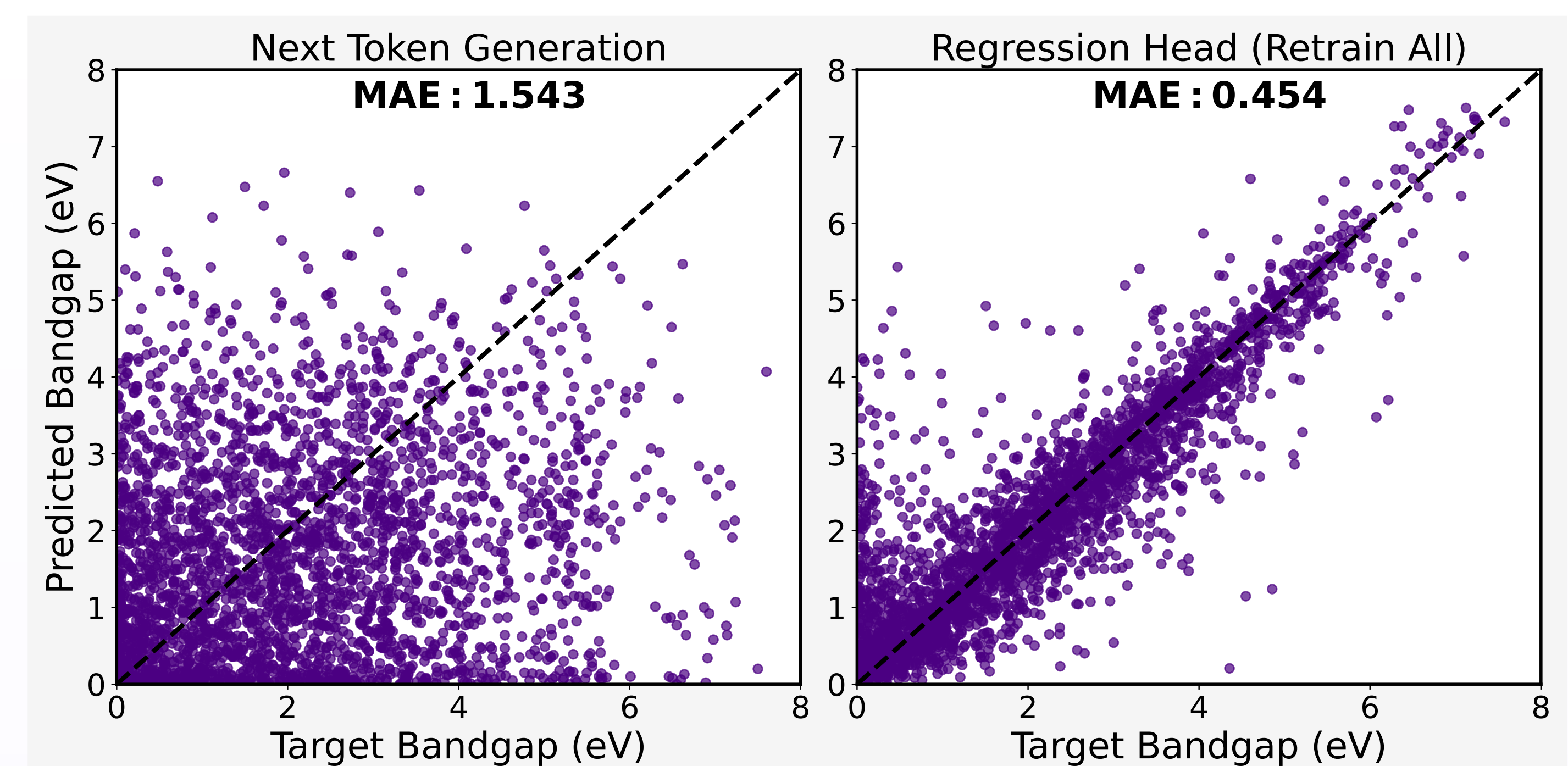


Figure 1. Model inference on bandgap predictions for next token generation (left) and regression head (right)

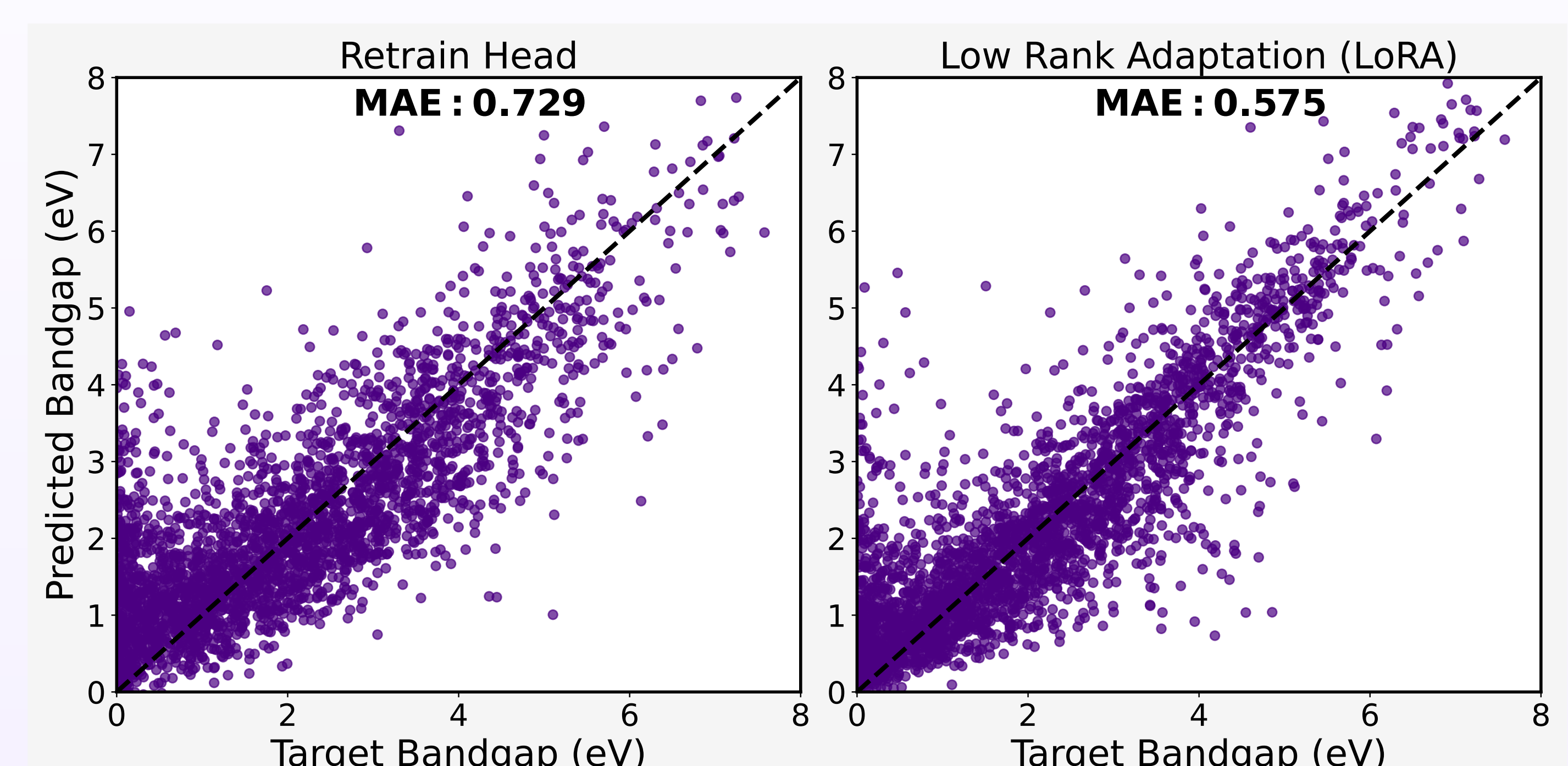


Figure 2. Evaluating regression when updating only model head parameters (left), compared to Low Rank Adaptation³ (right)

CONCLUSIONS

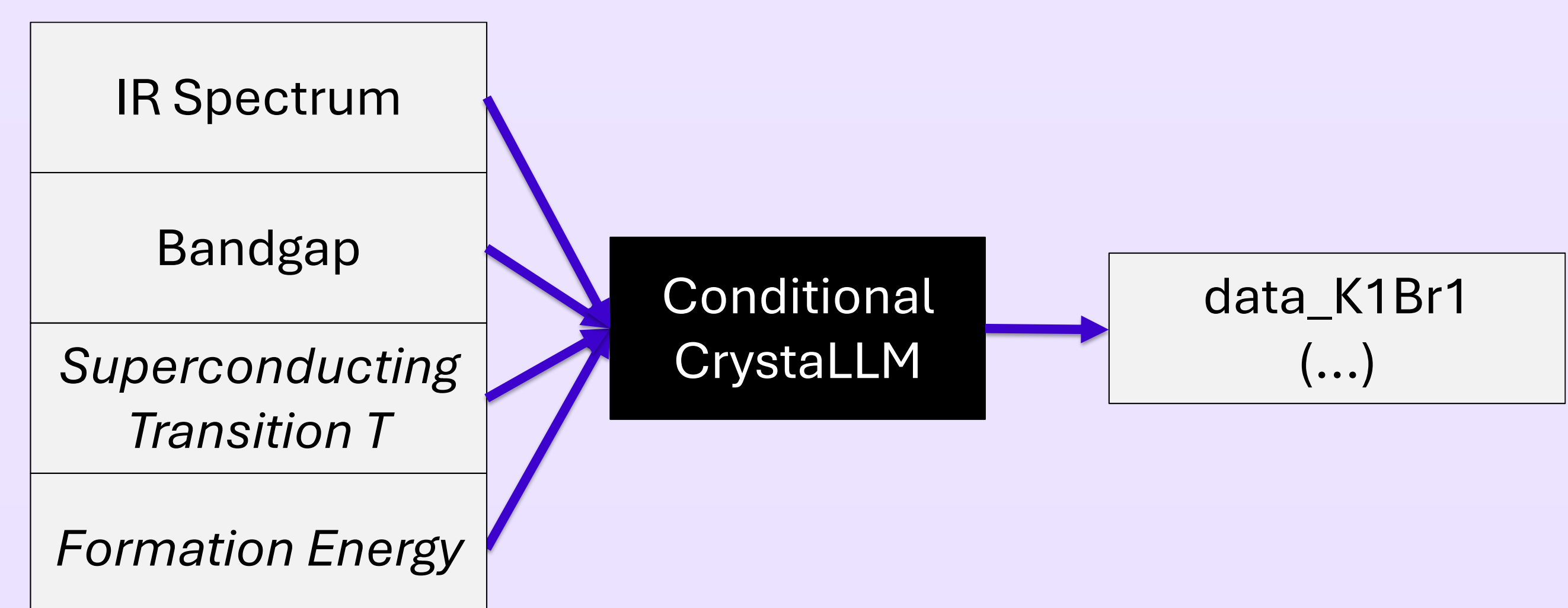
Task awareness is critical when data is scarce – a regression head logically outperforms next token generation.

Updating reduced sets of model parameters hinders prediction capabilities.

State-of-the-art transformers achieve MAEs of 0.2-0.3 for bandgap predictions on **material project** crystals:⁴

- CrystaLLM is small (~200 million vs 11 billion parameters for T5)⁵
- Model was only trained on sequences of 2048 tokens, here some predictions use up to 7000 tokens
- Model may not correctly leverage full CIF information for predictions

FUTURE WORK



Scheme 3. Retraining CrystaLLM for conditional generation

REFERENCES

- 1 L. M. Antunes, K. T. Butler and R. Grau-Crespo, 2024, preprint, arXiv:2307.04340, <http://arxiv.org/abs/2307.04340>.
- 2 A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder and K. A. Persson, APL Materials, 2013, 1, 011002.
- 3 E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang and W. Chen, arXiv, 2021, preprint, arXiv:arXiv:2106.09685, DOI: 10.48550/arXiv.2106.09685.
- 4 A. N. Rubungo, C. Arnold, B. P. Rand and A. B. Dieng, arXiv, 2023, preprint, arXiv:arXiv:2310.14029, <http://arxiv.org/abs/2310.14029>.
- 5 C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li and P. J. Liu, Journal of Machine Learning Research, 2020, 21, 1–67.
- 6 B. Courty, V. Schmidt, G. Kamal, M. Coutarel, B. Feld, J. Lecourt, L. Connell, et al., Zenodo, 2024, DOI: 10.5281/zenodo.11171501.

E-mail: cyprien.bone.24@ucl.ac.uk & k.t.butler@ucl.ac.uk